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Comparative Analysis of ARIMA and LSTM Models for Stock Price Prediction

Smit Anilkumar Panchal
Physics and Computer Science
Wilfrid Laurier University
Waterloo, ON, Canada
panc7220@mylaurier.ca

Lilatul Ferdouse
Physics and Computer Science
Wilfrid Laurier University
Waterloo, ON, Canada
lferdouse@wlu.ca

Ajmery Sultana
School of Computer Science and Technology
Algoma University
Brampton, ON, Canada
ajmery.sultana@algomau.ca

***Abstract*—Stock price prediction is crucial for informed investment decisions, enabling investors to maximize returns and manage risks effectively in the dynamic and complex world of financial markets. It also aids in portfolio management and financial planning by providing insights into future market movements and asset valuations.This study delves into the intriguing realm of stock price prediction using two models, Auto-Regressive Integrated Moving Average (ARIMA) and Long Short-Term Memory (LSTM) networks, leveraging the efficient market hypothesis framework. Analyzing historical market data for Apple, Google, and Tesla, ARIMA and LSTM models are independently developed to forecast closing stock values. The research compares the forecasting accuracy of each model through Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE) assessment, aiming to provide insights into their distinct strengths. The findings offer nuanced perspectives on the predictive performance of ARIMA and LSTM models in stock price behavior.**

***Index Terms*—Stock price prediction, financial markets, ARIMA, LSTM, forcasting accuracy.**

I. Introduction

Predicting stock prices has always been challenging, which has led to a constant examination of approaches ranging from conventional theories to cutting-edge technology. In the 1960s, Eugene Fama established the Efficient Market Hypothesis (EMH), which challenged the viability of precise forecasts by proposing that stock prices efficiently incorporate all available information. Later studies, on the other hand, focused on different strategies, such as technical analysis and more recently, the use of machine learning and deep learning.

The foundation of technical analysis was established by early pioneers such as John Magee and Robert D. Edwards, who introduced the idea of recognizing trends in stock price movements. This ground- breaking research was essential in comprehending market behavior since it paved the way for the widespread use of historical data patterns in predictive modeling [1].

In recent decades, a revolutionary shift occurred with the advent of machine learning. One of Marcos Lopez de Prado's achievements, particularly in "Advances in Financial Machine Learning" was highlighting the use of sophisticated algorithms to handle financial data efficiently. In terms of using computational models for predictive analytics in finance, this divergence from conventional statistical methods represents a crucial evolution [2].

Long Short-Term Memory (LSTM) networks are becoming a growing trend in predictive modeling because of their ability to capture complex temporal relationships. In contrast, Auto-Regressive Integrated Moving Average (ARIMA) models, which are well-established in the field of time series analysis, are beginning to be considered essential tools for predicting financial trends [3]. Rob J. Hyndman and George Athanasopoulos's research has been crucial in expanding the use of ARIMA, demonstrating the extent to which it can capture time-related patterns in datasets that include financial time series [4].

In this paper, ARIMA and LSTM networks, two popular models for stock price prediction are compared. The objective is to conduct an in-depth evaluation of the accuracy of both models in forecasting the closing stock prices of three significant tech giants: Apple, Google, and Tesla. Through a systematic comparison of results, the paper contributes significant value to the current discussion over the relative benefits and drawbacks of traditional and modern forecasting methods in the constantly evolving field of stock market predictions.

The rest of this paper is organized as follows. Section II describes the related work in the field of stock price prediction. Section III provides an overview of the background and description of ARIMA and LSTM model. Section IV presents experimental results. Finally, Section V concludes the paper.

II. Literature Review

This section focuses on related work in stock price prediction. Kumaravel and Soman delved into the efficacy of LSTM, Recurrent Neural Network (RNN), and Convolutional Neural Network (CNN) models for stock price prediction [5]. A comparative analysis of these neural network architectures was made easier by the study's use of a sliding window model. Financial time series data included complex temporal patterns that were best decoded by LSTM, which is well-known for its ability in capturing long-term dependencies.

In order to improve stock price prediction accuracy, Ding et al. advanced the discourse by combining sentiment analysis from financial news with deep learning, specifically LSTM

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	Date	Open	High	Low	Close	Adj Close	Volume
0	1980-12-12	0.128348	0.128906	0.128348	0.128348	0.099450	469033600
1	1980-12-15	0.122210	0.122210	0.121652	0.121652	0.094261	175884800
2	1980-12-16	0.113281	0.113281	0.112723	0.112723	0.087343	105728000

Fig. 1. Dataset of stocks

networks [6]. The study combined sentiment analysis extracted from financial news articles with LSTM networks. By incorporating another input into the deep learning model, this integration improved the predictive framework. The integration of sentiment analysis resulted in a significant improvement in prediction accuracy. This indicates external factors, specifically news sources' market sentiment, play a significant role in fine-tuning deep learning models' predictive abilities.

Johnson and Smith conducted a thorough analysis of ARIMA model with the goal of improving financial forecasting [7]. Their research demonstrated the simplicity and interpretability of ARIMA models, underscoring the continued value of conventional statistical methods for stock price prediction.

Furthermore, Wang and Chen made a significant contribution to the literature by delving deeply into the topic of long short-term memory networks' stock price prediction [8]. Their research, which was published in the Journal of Finance and Analytics, offered insightful information about how to use sophisticated neural network architectures to make accurate stock price predictions.

III. Methodology

This section covers pre-processing of stock datasets and details about two models, ARIMA and LSTM that are used for predicting stock prices. The LSTM model, a deep learning approach, and the ARIMA model, a conventional time series forecasting method, were chosen to capture different aspects of the temporal dependencies present in financial time series data.

A. Pre-processing of dataset

The Yahoo Finance stock historical dataset shown in figure 1, sourced directly from Yahoo Finance, typically comprises several essential columns providing insights into a stock's historical trading activity. The "Date" column marks each row with the corresponding trading day. "Open" represents the opening price at the start of the trading day, while "High" and "Low" denote the highest and lowest prices reached during the day, respectively. "Close" indicates the price at the end of the trading day, serving as the last traded price within regular market hours. "Adj Close" offers an adjusted closing price, accounting for events like dividends or stock splits, providing a more accurate reflection of the stock's value. "Volume" records the total number of shares traded on a given day, offering insights into market activity. Together, these columns furnish a comprehensive dataset for analyzing and understanding a stock's performance over time.

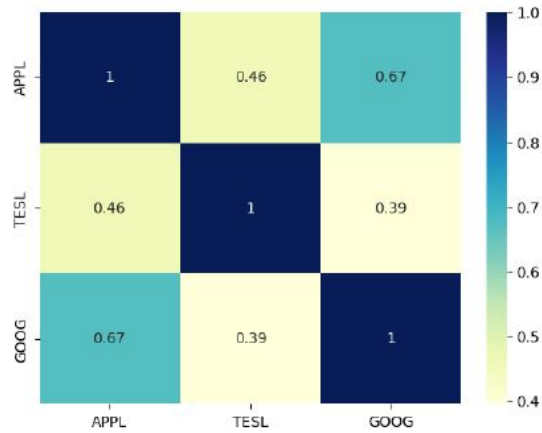

Fig. 2. Correlation between daily returns of stocks

The initial phase was dedicated to meticulous data preprocessing for the stocks Apple, Google and Tesla. The main objectives were to guarantee the integrity and consistency of the data for further evaluations. In addition to making data type changes, such as changing the "Date" column to datetime and the "Volume" column to float, null values were methodically eliminated. Following a chronological sorting of the datasets, records prior to January 1, 2017, were eliminated in order to align the analyses with particular temporal goals.

B. Exploratory Data Analysis

In the lifecycle of data-centric projects, exploratory data analysis (EDA) is a crucial stage that provides deep insights that serve as the cornerstone for well-informed decision-making. EDA acted as a compass for this research project, directing the investigation of underlying patterns, relationships, and anomalies in the Tesla, Apple, and Google stock datasets. identifying significant trends, recognizing possible outliers, and understanding the complex dynamics in the financial time series data are the main goals of EDA.

EDA visualizations, such as time series plots of closing prices and daily trading volumes, provide a tangible narrative that goes beyond simple numerical data. Calculated metrics that offer nuanced perspectives on relative fluctuations and stock price trends include moving averages and daily returns. The use of correlation analysis to uncover complex relationships between the daily returns of various stocks enhances our understanding of market dynamics. Figure 2 indicates that Apple and Google has the highest correlation in daily stock returns.

C. ARIMA model

The process of developing an ARIMA model for stock price prediction includes several key steps, including determining the ARIMA model's parameters, performing a logarithmic transformation, and testing for stationarity.

A statistical test called the Dickey-Fuller test is used to assess if a time series is stationary. The alternative hypothesis proposes that there is no unit root, indicating stationarity, in

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contrast to the null hypothesis, which states that the series has a unit root, indicating non-stationarity. The null hypothesis is typically accepted when the p-value is greater than 0.05, which suggests non-stationarity. The Dickey-Fuller test plays a critical role in the ARIMA model when determining whether to differentiate the series to achieve stationarity.

Following the stationarity assessment, the "Adj Close" prices are transformed using a logarithmic function to stabilize the time series variance. This transformation is especially helpful when working with data that has trends and fluctuating variances over time. This reduces the magnitudes of the values and lessens the upward trend in the series by taking the natural logarithm of the stock prices. The transformed data is then used for subsequent modeling steps.

Based on the time series data that has been pre-processed, the ARIMA model is utilized to forecast future stock prices. ARIMA is a moving average, differencing, and auto-regression time series forecasting model. The order of the auto-regressive, integrated, and moving average components is represented by the ARIMA model's parameters, which are indicated by the letters θ, Φ, and q [9]. The ARIMA model can be expressed as:

$$\text{Y(t)} = \theta_1 * Y(t-1) + \theta_2 * Y(t-2) + ... + \theta_q * Y(t-q) + \epsilon, \quad (1)$$

where $Y(t)$ is the observed time series at time t, θ represents auto-regressive parameters, ϵ is the white noise error term which is calculated by the following equation

$$\epsilon = \epsilon(t) - \Phi_1 * \epsilon(t-1) - \Phi_2 * \epsilon(t-2) - ... - \Phi_q * \epsilon(t-q), \quad (2)$$

where Φ denotes the moving average parameters. The model's mathematical simplicity makes it computationally efficient and interpretable.

By automatically choosing the best values based on the supplied time series data, the auto-ARIMA function is used to determine these parameters. Next, future stock prices are predicted by fitting the ARIMA model to the training set.

In conclusion, the methodology includes using the Dickey-Fuller test to test for stationarity, the logarithmic transformation to stabilize variance, and the ARIMA model to predict stock prices. By taking these steps together, a predictive model that can identify and predict trends in stock price movements is created.

D. LSTM model

The LSTM neural networks have achieved prominence in stock price prediction due to their ability to capture and learn intricate temporal dependencies present in time series data. For tasks involving historical stock prices, where the ordering of data points is critical, LSTMs perform exceptionally well in modeling sequences, in contrast to traditional feed forward networks. Given the complex patterns and trends present in financial time series data, LSTM architecture's capacity to retain and selectively update information over extended sequences makes them an effective tool for stock price prediction. LSTMs are a preferred choice for stock market forecasting tasks due to their flexibility in capturing both short-term fluctuations and long-term patterns [10].

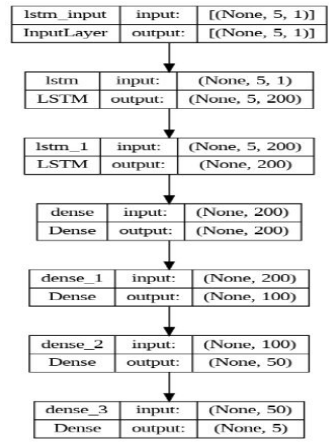

Fig. 3. LSTM Architecture

The LSTM model is built with the Keras Sequential API to predict stock prices, leveraging the temporal dependencies in time series data. As shown in figure 3, the input layer of the model begins with sequences of length 5, which correspond to historical stock prices. Long-term dependencies are captured using two LSTM layers, each with 200 units and Leaky ReLU activation. Sequences are returned to the next layer by the initial LSTM layer. Three densely connected layers (200, 100, and 50) with decreasing units process learned features after the LSTM layers. Predictions for the next 5 days are produced by the last dense layer, which has 5 units and activates the leaky ReLU.

A personalized learning rate schedule is used to optimize training. For the first 150 epochs, the schedule increases the learning rate linearly. Learning rate is adjusted by a decaying exponential function between epoch 150 and 400. A lower fixed learning rate ($10^{-}6$) is set after epoch 400. Through improved convergence and performance, this dynamic adjustment aids in the model's adaptation to various training phases.

A popular option for deep learning tasks, the Adam optimizer is used to compile the LSTM model. The loss function used to calculate the difference between expected and actual values is Mean Squared Error (MSE). A learning rate scheduler modifies the learning rate as the training progresses over a thousand epochs. Performance on the training and validation sets is tracked to evaluate the generalization capacity of the model.

During the testing period, the model forecasts closing stock prices after training. Performance metrics, including Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE), are computed by comparing the predicted values with the actual stock prices. The model's predictive accuracy over several time steps is quantitatively evaluated by these metrics.

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TABLE I
RMSE AND MAPE OF STOCKS USING ARIMA MODEL

ARIMA		
Stock	RMSE	MAPE
AAPL	13.60	0.064
GOOOG	7.15	0.048
TSLA	23.98	0.243

TABLE II
RMSE AND MAPE OF STOCKS USING LSTM MODEL

LSTM		
Stock	RMSE	MAPE
AAPL	5.52	0.024
GOOOG	4.66	0.028
TSLA	12.38	0.039

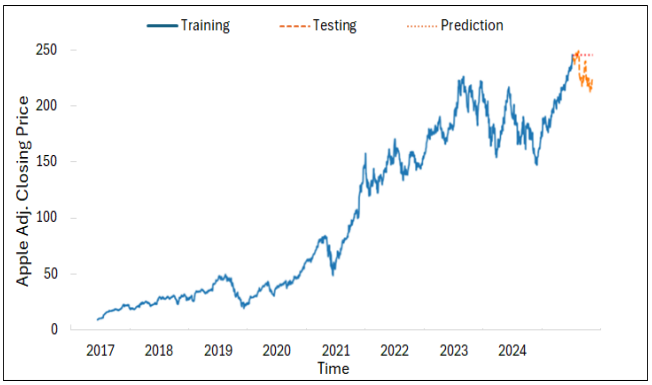

Fig. 4. ARIMA model prediction graph of AAPL

IV. EXPERIMENT AND RESULTS

Both ARIMA and LSTM models were used in our experimental evaluation of stock price prediction for Apple (AAPL), Tesla (TSLA), and Google (GOOG). With distinct RMSE values for every stock, the ARIMA model demonstrated an adequate degree of performance. For example as shown in the table I, the RMSE of 13.6 was obtained for Apple using the ARIMA model, 23.98 for Tesla, and 7.15 for Google. It is imperative to acknowledge that the ARIMA predictions were associated to MAPE values, which suggests a certain degree of imprecision (for example, Tesla's MAPE was 0.24).

The LSTM models, on the other hand, demonstrated their superiority in capturing complex patterns and non-linear correlations inherent in stock price movements by consistently outperforming ARIMA across all stocks. When compared to ARIMA, the LSTM models produced noticeably lower RMSE values. For instance, as shown in the table II, the RMSE of the LSTM for Apple was 5.52, for Tesla it was 12.38, and for Google it was 4.66. The matching MAPE values, which ranged from 0.024 to 0.039, were extremely low. These findings highlight how much more accurate and dependable LSTM is than the conventional ARIMA model for predicting stock prices.

The prediction graphs in figure 5, 7 and 9 when inspected visually, demonstrate how accurate LSTM is; aligning closely

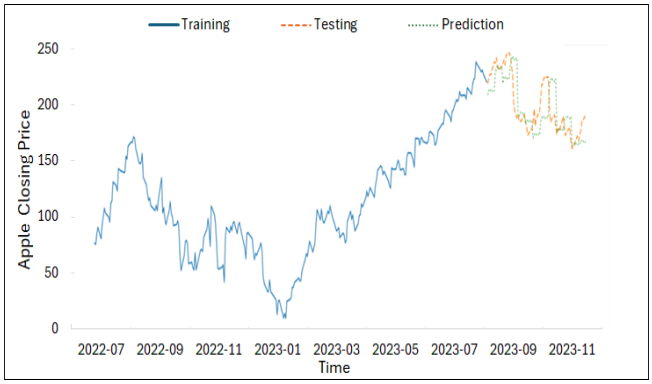

Fig. 5. LSTM model prediction graph of AAPL

Fig. 6. ARIMA model prediction graph of TSLA

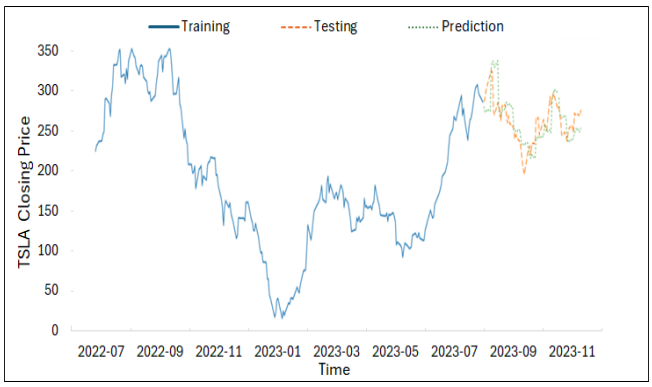

Fig. 7. LSTM model prediction graph of TSLA

to the actual stock prices and capturing even the most complex variations. On the other hand, as shown figure 4, 6 and 8, ARIMA predictions show a latency, particularly in times of increased volatility. This visual alignment validates the numerical results and shows that, in comparison to the more inflexible ARIMA model, LSTM is better able to adjust to the dynamic nature of stock prices.

The intrinsic properties of stock market data are the reason behind LSTM's better performance than ARIMA. It is well known that stock values exhibit non-linear patterns, abrupt changes, and dynamic reliance on a variety of factors. While

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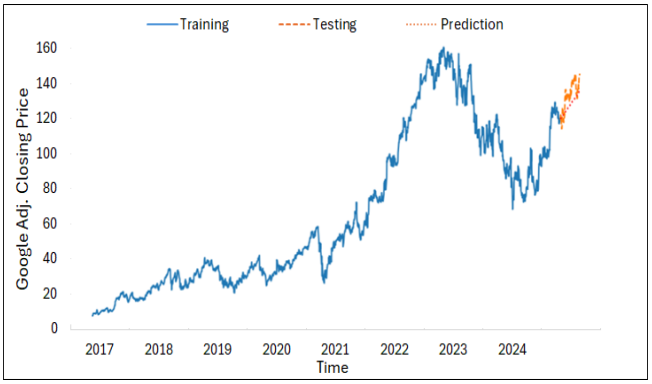

Fig. 8. ARIMA model prediction graph of GOOG

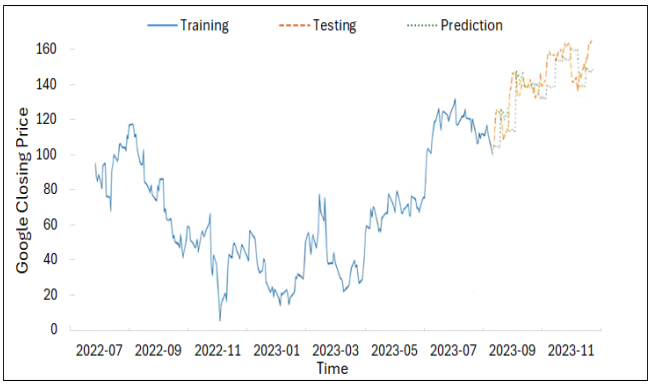

Fig. 9. LSTM model prediction graph of Tesla

LSTM is excellent at identifying intricate patterns and adjusting to changing market conditions, ARIMA is dependent on linear relationships and has trouble with sudden fluctuations in volatility. Moreover, the flexibility of LSTM in managing various characteristics and integrating extraneous elements enhances its efficacy.

The use of ARIMA in stock price prediction highlights some of its limitations, including its susceptibility to constant variance assumptions and difficulties in accommodating nonlinear patterns. LSTM is a better option for stock price prediction because of its capacity to automatically learn relevant data, adjust to changing market conditions, and capture long-term dependencies. As a result, the experiments highlight the important benefits of LSTM for stock price prediction and highlight the drawbacks of ARIMA for managing the complex dynamics of financial markets.

V. Conclusion

To conclude, this work examined the ARIMA and LSTM models for predicting the stock prices of Google, Apple, and Tesla. The accuracy of the ARIMA models varied, with RMSE ranging from 7.15 to 23.9. However, their performance was restricted in capturing intricate patterns. Conversely, the LSTM models demonstrated better predictive power than their ARIMA counterparts for every stock that was evaluated. Lower RMSE values were produced by the LSTM models, highlighting their efficacy in capturing the complex patterns and temporal relationships seen in stock price fluctuations. In particular, the LSTM model for Google achieved 4.66 RMSE, Tesla 12.38 RMSE, and Apple 5.52 RMSE, demonstrating significant accuracy gains over the ARIMA models. The disparity in outcomes highlights the constraints of conventional time-series analysis methods such as ARIMA when faced with the intricacy of stock price data. The LSTM models seem to be a more reliable method for predicting stock prices because of their capacity to identify long-term dependencies and adjust to nonlinear patterns. Our results underline how crucial it is to use cutting-edge machine learning methods, especially LSTM models, to anticipate stock prices in dynamic financial markets with more accuracy and dependability.

References

[1] R. D. Edwards and J. Magee, “Technical Analysis of Stock Trends”, CRC Press, 2007.
[2] M. Lopez de Prado, “Advances in Financial Machine Learning”, John Wiley & Sons, 2018.
[3] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, “Time Series Analysis: Forecasting and Control,” Wiley 2015.
[4] R. J. Hyndman and G. Athanasopoulos, “Forecasting: principles and practice,” OTexts, 2018.
[5] N. Kumaravel and K. P. Soman, “Stock price prediction using LSTM, RNN and CNN-sliding window model,” IEEE International Conference on Advances in Computing, Communications and Informatics (ICACCI), pp. 1647-1652, 2018.
[6] X. Ding, Y. Zhang, T. Liu, and Q. Duan, “Stock price prediction using deep learning and sentiment analysis,” Proceedings of the 26th International Joint Conference on Artificial Intelligence, pp. 2597-2603, AAAI Press, 2017.
[7] A. Johnson and B. Smith, “Enhancing Financial Forecasting: A Comprehensive Review of ARIMA Models,” Journal of Financial Analysis, vol. 45, no. 2, pp. 123-145, 2019.
[8] M. Wang and Q. Chen, “Predicting Stock Prices with Long Short-Term Memory Networks: A Comprehensive Study”, Journal of Finance and Analytics, vol. 12, no. 3, pp. 211-230, 2020.
[9] S. Hochreiter, and J. Schmidhuber, “Long short-term memory,” Neural Computation, 9(8), 1735-1780, 1997.
[10] D. T. Nguyen, D. T. Nguyen, and H. Q. Nguyen, “A deep learning framework for financial time series using stacked autoencoders and long-short term memory,” IEEE 2nd International Conference on Cloud Computing and Big Data Analysis (ICCCBDA), pp. 475-480, 2017.

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